



Hospital Engineering and Management

Lecture 1: Hospitals as Complex Engineering and Management Systems

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Course Code: UBM702

Learning Objectives

By the end of this lecture, students should be able to:

- ▶ define hospital engineering and hospital management;
- ▶ explain why a hospital is considered a complex socio-technical system;
- ▶ identify the major clinical, support and engineering services in a hospital;
- ▶ describe the role of engineers in hospital planning, operation and safety;
- ▶ recognize the importance of reliability, safety, infection control and patient flow;
- ▶ relate engineering decisions to patient care and healthcare quality.

Evaluation Scheme

- ▶ Mid Semester Test 30 Marks
- ▶ End Semester Test 40 Marks
- ▶ Sessional 30 Marks

Sessional Includes

1. Quiz 10 Marks
2. Mini Project (Presentation, Report and Involvement 5, 10 and 5 Marks each)

Opening Question: What Keeps a Hospital Running?

Think About It

A surgeon may perform an operation, but what other systems must work correctly for the surgery to be safe and successful?

- ▶ uninterrupted electrical power;
- ▶ medical gases and vacuum;
- ▶ sterile instruments;
- ▶ infection-control procedures;
- ▶ ventilation and air-conditioning;
- ▶ diagnostic and monitoring equipment;
- ▶ trained personnel;
- ▶ safe water and waste disposal.

A hospital is not merely a building; it is an integrated system.

What Is Hospital Engineering?

Definition

Hospital engineering is the application of engineering principles to the planning, design, installation, operation, maintenance and safety of healthcare facilities.

It includes:

- ▶ hospital buildings and layouts;
- ▶ electrical infrastructure;
- ▶ HVAC systems;
- ▶ water and sanitation;
- ▶ medical gas systems;
- ▶ medical equipment management;
- ▶ fire and emergency systems;
- ▶ energy and facility management.

Engineering knowledge + Clinical requirements = Safe healthcare infrastructure

What Is Hospital Management?

Hospital management coordinates people, processes, technology and resources to provide safe and efficient healthcare services.

- ▶ planning and organization of hospital services;
- ▶ human-resource and workforce management;
- ▶ patient admission, transfer and discharge;
- ▶ medical records and information management;
- ▶ finance, inventory and procurement;
- ▶ quality assurance and regulatory compliance;
- ▶ coordination of clinical and support departments;
- ▶ emergency, risk and disaster management.

**Engineering ensures that individual systems work;
management ensures that all systems work together.**

Why Study Hospital Engineering and Management?

Hospitals must operate continuously despite:

- ▶ equipment failures;
- ▶ power interruptions;
- ▶ high and variable patient loads;
- ▶ emergency situations;
- ▶ infection risks;
- ▶ limited resources;
- ▶ changing medical technologies;
- ▶ regulatory requirements.

Central Question

How can a hospital provide safe, reliable and efficient healthcare under uncertain and demanding conditions?

Engineering Response

Design systems with appropriate reliability, redundancy, maintainability, monitoring and emergency backup.

Hospital as a Socio-Technical System

A hospital combines human, technological, organizational and infrastructural elements.

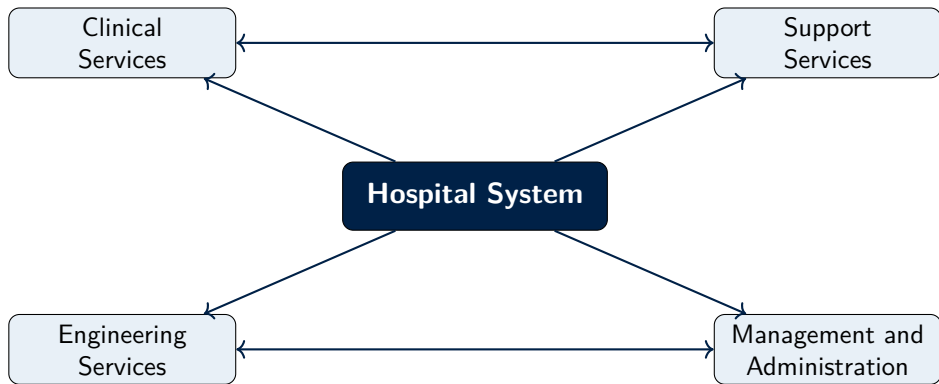
$$\text{People} + \text{Technology} + \text{Infrastructure} + \text{Processes} = \text{Healthcare Delivery}$$

- ▶ **People:** doctors, nurses, engineers, technicians, administrators, patients and attendants;
- ▶ **Technology:** medical equipment, information systems, sensors and communication networks;
- ▶ **Infrastructure:** buildings, electricity, HVAC, water, sanitation and medical gases;
- ▶ **Processes:** diagnosis, treatment, sterilization, documentation and patient movement.

Important

Failure of any one element can affect patient safety, service quality and hospital performance.

Hospital as a System of Interconnected Services



No hospital department can function effectively in isolation.

Major Functional Areas of a Hospital

Clinical Services

- ▶ outpatient department;
- ▶ emergency department;
- ▶ inpatient wards;
- ▶ intensive care units;
- ▶ operation theatres;
- ▶ radiology and imaging;
- ▶ laboratories;
- ▶ pharmacy.

Support Services

- ▶ central sterile supply department;
- ▶ laundry and linen services;
- ▶ dietary and food services;
- ▶ housekeeping;
- ▶ medical records;
- ▶ security and transportation;
- ▶ stores and procurement;
- ▶ waste-management services.

Major Engineering Systems in a Hospital

- ▶ **Electrical system:** power distribution, backup generators, UPS, earthing and lighting;
- ▶ **HVAC system:** temperature, humidity, ventilation, filtration and pressure control;
- ▶ **Medical gas system:** oxygen, medical air, nitrous oxide, carbon dioxide and vacuum;
- ▶ **Water and sanitation:** potable water, hot water, drainage, sewage and water treatment;
- ▶ **Fire-protection system:** alarms, detectors, extinguishers, sprinklers and evacuation routes;
- ▶ **Communication system:** nurse call, telephone, networking, public address and emergency alerts;
- ▶ **Medical equipment system:** installation, calibration, maintenance and electrical-safety testing;
- ▶ **Building-management system:** centralized monitoring of utilities, alarms and energy use.

Patient Flow and Hospital Design

A good hospital layout supports safe and efficient movement of:

- ▶ patients;
- ▶ doctors and hospital staff;
- ▶ visitors and attendants;
- ▶ emergency vehicles;
- ▶ medicines and sterile materials;
- ▶ food, linen and supplies;
- ▶ laboratory samples;
- ▶ biomedical waste.

Arrival → Registration → Diagnosis → Treatment → Recovery → Discharge

Important Design Principle

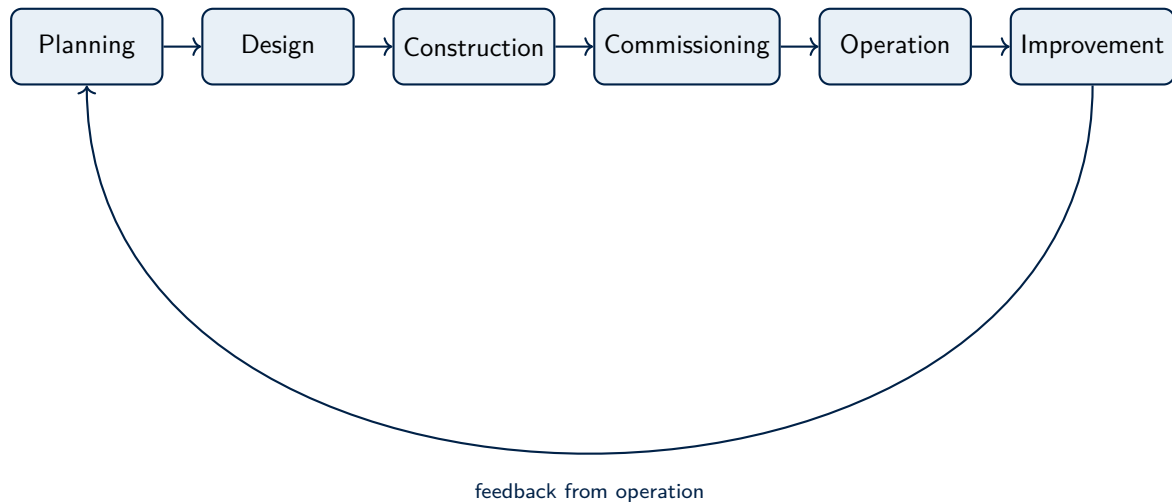
Clean, sterile and contaminated flows should be separated wherever possible.

Role of a Hospital Engineer

A hospital engineer contributes throughout the complete facility life cycle.

1. **Planning:** estimate capacity, space, utilities and technical requirements;
2. **Design:** coordinate architectural, clinical and engineering requirements;
3. **Installation:** supervise equipment installation, testing and commissioning;
4. **Operation:** ensure continuous availability of essential services;
5. **Maintenance:** conduct preventive, predictive and corrective maintenance;
6. **Safety:** reduce electrical, mechanical, fire and environmental risks;
7. **Improvement:** optimize energy use, workflow, reliability and cost.

Hospital Facility Life Cycle



Reliability and Redundancy

Some hospital systems are safety-critical and cannot be allowed to fail.

Safety-Critical Examples

- ▶ power supply to intensive care units;
- ▶ oxygen supply in operation theatres;
- ▶ ventilators and patient monitors;
- ▶ ventilation in isolation rooms;
- ▶ fire-detection systems.

Typical Backup Provisions

- ▶ diesel generator;
- ▶ uninterruptible power supply;
- ▶ reserve gas cylinders;
- ▶ duplicated pumps and compressors;
- ▶ emergency procedures.

Reliability = Good Design + Redundancy + Maintenance + Monitoring

Safety in Hospitals

Hospital safety includes several interconnected areas:

- ▶ patient safety;
- ▶ electrical safety;
- ▶ medical-equipment safety;
- ▶ fire and life safety;
- ▶ radiation safety;
- ▶ infection prevention and control;
- ▶ occupational safety;
- ▶ chemical and biological safety;
- ▶ biomedical-waste safety;
- ▶ disaster preparedness.

Key Message

A hospital must remain safe during normal operation, equipment failure, natural disasters and mass-casualty emergencies.

Case Study: Power Failure in an ICU

Suppose the main electrical supply fails while several patients are connected to ventilators and monitoring equipment.

Discussion Questions

1. Which systems must continue operating immediately?
2. What backup systems should be available?
3. How quickly should emergency power become available?
4. Which hospital personnel should receive an alarm?
5. What preventive measures can reduce the risk?

Engineering Lesson

Hospital systems must be designed by anticipating possible failures and providing safe fallback mechanisms.

Stakeholders in Hospital Planning

Hospital planning requires coordination among:

- ▶ hospital owners;
- ▶ administrators;
- ▶ doctors and nurses;
- ▶ patients and attendants;
- ▶ architects;
- ▶ civil engineers;
- ▶ electrical engineers;
- ▶ mechanical engineers;
- ▶ biomedical engineers;
- ▶ infection-control specialists;
- ▶ safety and quality teams;
- ▶ regulatory authorities.

Successful hospital planning is multidisciplinary.

Course Roadmap

During this course, we will study:

1. planning and organization of hospitals;
2. clinical services and functional departments;
3. design of hospital engineering services;
4. maintenance and management of hospital infrastructure;
5. support services and medical-record systems;
6. fire safety and disaster management;
7. infection prevention and control;
8. biomedical-waste management;
9. team-based hospital engineering projects.

Classroom Activity

Activity: Identify the Hidden Engineering

Select any one hospital area:

- ▶ operation theatre;
- ▶ intensive care unit;
- ▶ emergency department;
- ▶ radiology department;
- ▶ clinical laboratory.

For the selected area, identify:

1. major equipment;
2. required utilities;
3. possible hazards;
4. necessary backup systems;
5. important maintenance activities.

Summary

- ▶ A hospital is a complex and safety-critical socio-technical system.
- ▶ Hospital engineering integrates infrastructure, utilities, equipment and safety.
- ▶ Hospital management coordinates people, processes and resources.
- ▶ Clinical and support departments depend on reliable engineering services.
- ▶ Good planning improves patient safety, efficiency and quality of care.
- ▶ Engineers contribute throughout the complete hospital life cycle.

Safe hospitals are engineered, managed and continuously improved.

Assignment for the Next Lecture

Visit a hospital physically or examine the layout and services of a hospital through its website or published information.

Prepare a one-page note containing:

1. the type and approximate size of the hospital;
2. major clinical departments;
3. major support services;
4. three important engineering systems;
5. one safety-related observation;
6. one suggestion for improving hospital operation.

Submission Format

One page, individually prepared, with a simple block diagram or departmental layout wherever possible.

Suggested Reading



S. Goel et al.,
Textbook of Hospital Administration: A Problem-Solving Approach,
Elsevier India.



G. D. Kunders,
Hospitals: Facilities Planning and Management,
Tata McGraw-Hill.



B. M. Sakharkar,
Principles of Hospital Administration and Planning,
Jaypee Brothers Medical Publishers.

Next Lecture: Planning and Organization of Hospitals

Thank You

Questions and Discussion

